

Multi-Sensor TOA-Based 2D Target Localization and Tracking

A Technical Interpretation of the MTH407 Term Project

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Abstract

We design, implement, and analyze a 2D target tracking system that fuses time-of-arrival (TOA) measurements from $N \in \{2, 3, 4\}$ stationary sensors with known positions. A least-squares (LSE) position estimator initializes an Extended Kalman Filter built on a constant-velocity (CV) motion model with white-acceleration process noise. We sweep six (*sensor count, geometry*) cells spanning favorable and degraded configurations, run a 100-trial Monte Carlo per cell, decompose the position RMSE into smooth-segment and turn-window components, and run two stress tests: a study of the warm-start prior's role in the $N=3$ collinear ambiguity, and a σ_a sensitivity sweep that exposes the CV-EKF's process-noise tuning trade-off. The results confirm the expected geometry-dilution scaling ($\text{RMSE} \sim \sigma_r / \sqrt{N}$) on smooth segments, identify unmodelled-maneuver lag as the dominant contributor to the blended RMSE metric, and demonstrate that the apparent robustness of the collinear geometry in the default configuration is wholly due to the warm-start prior breaking the mirror-image symmetry.

1 Introduction

The course specification (MTH407 dönem projesi) requires a 2D surveillance system in which time-based measurements from multiple known-position sensors are used to estimate and track the position of a moving target. The three required components are: a measurement-domain simulator (TOA or TDOA), an LSE-based estimator that produces an initial state vector and covariance, and an EKF-based tracker. The required deliverable is an analysis of how *sensor geometry* affects tracking accuracy.

This report describes the choices made and the resulting performance. Among the spec's options we use the TOA model rather than TDOA, fix the target trajectory to a constant-speed zig-zag with four sharp heading changes, exercise six geometry cells (two configurations each at $N \in \{2, 3, 4\}$), and quantify performance both deterministically and via a 100-trial Monte Carlo per cell. Beyond the basic comparison we add two analyses that the results suggested were necessary: a stress test of the $N=3$ collinear ambiguity, and a sensitivity sweep of the EKF's process-noise tuning parameter σ_a .

Implementation. The full pipeline is implemented in Python (NumPy / SciPy / Matplotlib) and distributed as the public repository <https://github.com/selcuko/mth407-toa-target-tracking>. All figures and tables in this document are reproduced from the Monte Carlo runs in that repository.

2 System Model

2.1 Geometry and Notation

Sensors $i = 1, \dots, N$ have known stationary positions $\mathbf{s}_i \in \mathbb{R}^2$. The target has unknown position $\mathbf{p}^k \in \mathbb{R}^2$ and velocity $\mathbf{v}^k \in \mathbb{R}^2$ at discrete time $t_k = kT$, with sample period $T = 0.5$ s. We collect the kinematic state into $\mathbf{x}^k = [\mathbf{p}^{k\top} \ \mathbf{v}^{k\top}]^\top \in \mathbb{R}^4$.

The surveillance area is a $10 \text{ km} \times 10 \text{ km}$ square, the target speed is fixed at $\|\mathbf{v}^k\| = 100 \text{ m/s}$, and the simulation runs for 60 s, yielding $K = 121$ time steps.

2.2 Ground-Truth Trajectory

The ground-truth trajectory is a five-leg, constant-speed, piecewise-linear zig-zag with heading sequence $\theta = (30^\circ, -30^\circ, 50^\circ, -50^\circ, 30^\circ)$ and starting point $(2000, 5000) \text{ m}$. Each leg lasts 12 s (24 steps), so heading changes occur at steps $\{24, 48, 72, 96\}$. These four turns are the core difficulty for the tracker: the EKF's CV motion model treats them as instantaneous, unmodelled accelerations.

2.3 TOA Measurement Model

Each sensor records the arrival time z_i^k of a signal emitted by the target at the known time t_k :

$$z_i^k = t_k + \frac{\|\mathbf{p}^k - \mathbf{s}_i\|}{c} + n_i^k, \quad n_i^k \sim \mathcal{N}(0, \sigma_t^2), \quad (1)$$

with signal speed $c = 3 \cdot 10^8 \text{ m/s}$ and time-measurement standard deviation $\sigma_t = 10^{-8} \text{ s}$. Internally we simulate the additive noise in the time domain, faithful to the physics, and convert to the *range domain* only at the LSE/EKF boundary:

$$r_i^k = c(z_i^k - t_k) = \|\mathbf{p}^k - \mathbf{s}_i\| + \nu_i^k, \quad \nu_i^k \sim \mathcal{N}(0, \sigma_r^2), \quad (2)$$

with effective range-noise standard deviation $\sigma_r = c\sigma_t = 3 \text{ m}$. The measurement noise is independent across sensors and across time, so the per-step measurement covariance is $\mathbf{R} = \sigma_r^2 \mathbf{I}_N$.

Why simulate in the time domain. Adding range-domain noise directly would give the same numerical result for a single linear conversion, but framing the simulator in the time domain preserves the physical interpretation and makes future extensions (clock biases, asynchronous sensors, TDOA) straightforward.

2.4 Target Dynamics (CV Model)

For the EKF we adopt a constant-velocity model in 2D:

$$\mathbf{x}^{k+1} = \mathbf{F}(T) \mathbf{x}^k + \mathbf{w}^k, \quad \mathbf{w}^k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad (3)$$

with

$$\mathbf{F}(T) = \begin{bmatrix} \mathbf{I}_2 & T\mathbf{I}_2 \\ \mathbf{0} & \mathbf{I}_2 \end{bmatrix}, \quad \mathbf{Q} = \sigma_a^2 \begin{bmatrix} \frac{T^4}{4}\mathbf{I}_2 & \frac{T^3}{2}\mathbf{I}_2 \\ \frac{T^3}{2}\mathbf{I}_2 & T^2\mathbf{I}_2 \end{bmatrix}. \quad (4)$$

The discrete white-acceleration form of $\mathbf{Q}$ models the velocity as a continuous-time random walk; σ_a has units of acceleration (m/s^2) and quantifies how much velocity change between samples the filter is willing to absorb. We use $\sigma_a = 10 \text{ m/s}^2$ by default and study the effect of varying it in §5.5. Sharp heading changes in the ground-truth trajectory are *not* explicitly modelled; they are absorbed by the process noise term and produce transient prediction errors that are the main story of §5.3.

3 Estimators

3.1 Least-Squares Position Estimation ($N \geq 3$)

For $N \geq 3$ sensors the per-step position estimator is iterative Gauss-Newton on the non-linear range residual. Stack the N range measurements at step k as $\mathbf{r}^k \in \mathbb{R}^N$; the residual at iterate $\mathbf{p}^{(j)}$ is

$$\boldsymbol{\rho}(\mathbf{p}^{(j)}) = \mathbf{r}^k - \mathbf{h}(\mathbf{p}^{(j)}), \quad h_i(\mathbf{p}) = \|\mathbf{p} - \mathbf{s}_i\|, \quad (5)$$

and the Jacobian of $\mathbf{h}$ is

$$\mathbf{H}(\mathbf{p}) = \begin{bmatrix} \mathbf{u}_1^\top \\ \vdots \\ \mathbf{u}_N^\top \end{bmatrix}, \quad \mathbf{u}_i = \frac{\mathbf{p} - \mathbf{s}_i}{\|\mathbf{p} - \mathbf{s}_i\|}. \quad (6)$$

That is, the i -th row of $\mathbf{H}$ is the unit line-of-sight vector from sensor i to the target. The Gauss-Newton step is

$$\mathbf{p}^{(j+1)} = \mathbf{p}^{(j)} + (\mathbf{H}^\top \mathbf{H})^{-1} \mathbf{H}^\top \boldsymbol{\rho}(\mathbf{p}^{(j)}), \quad (7)$$

with $\mathbf{H}^+$ used in place of $(\mathbf{H}^\top \mathbf{H})^{-1} \mathbf{H}^\top$ when the latter is ill-conditioned (this is the case the collinear geometry is supposed to expose). The iteration is initialised either at the centroid of the sensors (cold start) or at the previous step's estimate (warm start), and terminates when $\|\Delta \mathbf{p}\| < 10^{-3}$ m or after 30 iterations.

After convergence, the position covariance is approximated by the local Cramér–Rao–style bound

$$\mathbf{P}_p = \sigma_r^2 (\mathbf{H}^\top \mathbf{H})^{-1}, \quad (8)$$

evaluated at the estimate. Equation (8) makes the *geometric dilution of precision* (GDOP) explicit: the smaller the spread of the unit vectors $\mathbf{u}_i$, the worse-conditioned $\mathbf{H}^\top \mathbf{H}$ becomes, and the larger the position uncertainty for the same per-range σ_r .

3.2 LSE for $N = 2$ (Two-Circle Intersection)

For $N = 2$ the range residuals over-determine no parameters; instead the two range constraints define two circles, which intersect in $\{0, 1, 2\}$ points. Let $d = \|\mathbf{s}_2 - \mathbf{s}_1\|$ and define $\mathbf{e}_x = (\mathbf{s}_2 - \mathbf{s}_1)/d$ and a right-handed perpendicular $\mathbf{e}_y$. Define

$$a = \frac{r_1^2 - r_2^2 + d^2}{2d}, \quad h = \sqrt{\max(r_1^2 - a^2, 0)}. \quad (9)$$

The two candidate intersections are $\mathbf{p}_\pm = \mathbf{s}_1 + a\mathbf{e}_x \pm h\mathbf{e}_y$. We pick the one closer to a coarse prior $\mathbf{p}_{\text{prior}}$ (provided externally, e.g. from rough operational knowledge of the target's expected start region). The covariance is approximated by (8) evaluated at the chosen candidate. This estimator is fundamentally ambiguous and depends on the prior; the analogous degeneracy for $N=3$ collinear is the subject of §5.4.

3.3 State-Vector Initialization

The initial 4-D state $\hat{\mathbf{x}}_0$ is constructed from LSE position estimates at the first two time steps:

$$\hat{\mathbf{p}}_0 = \text{LSE}(\mathbf{r}^0), \quad \hat{\mathbf{p}}_1 = \text{LSE}(\mathbf{r}^1), \quad \hat{\mathbf{v}}_0 = \frac{\hat{\mathbf{p}}_1 - \hat{\mathbf{p}}_0}{T}. \quad (10)$$

The covariance $\mathbf{P}_0$ inherits position-block from (8), the velocity block from finite-difference propagation $\mathbf{P}_v \approx (\mathbf{P}_{p,0} + \mathbf{P}_{p,1})/T^2$, and the position–velocity cross-block from $\mathbf{P}_{pv} \approx -\mathbf{P}_{p,0}/T$, which follows from $\text{cov}(\hat{\mathbf{p}}_0, \hat{\mathbf{v}}_0) = -\text{cov}(\hat{\mathbf{p}}_0, \hat{\mathbf{p}}_0)/T$ when $\mathbf{r}^0$ and $\mathbf{r}^1$ are independent.

3.4 Constant-Velocity Extended Kalman Filter

The EKF runs the standard predict/update loop with the CV model (4) and the per-sensor range measurement (2). The prediction is

$$\hat{\mathbf{x}}^{k|k-1} = \mathbf{F} \hat{\mathbf{x}}^{k-1|k-1}, \quad \mathbf{P}^{k|k-1} = \mathbf{F} \mathbf{P}^{k-1|k-1} \mathbf{F}^\top + \mathbf{Q}. \quad (11)$$

For the update we stack all N range innovations into a single vector and form the $(N \times 4)$ Jacobian by extending each row of (6) with two zeros (since the range function does not depend on velocity):

$$\mathbf{H}^k = \begin{bmatrix} \mathbf{u}_1^\top & \mathbf{0}_{1 \times 2} \\ \vdots & \vdots \\ \mathbf{u}_N^\top & \mathbf{0}_{1 \times 2} \end{bmatrix}, \quad \mathbf{u}_i = \frac{\hat{\mathbf{p}}^{k|k-1} - \mathbf{s}_i}{\|\hat{\mathbf{p}}^{k|k-1} - \mathbf{s}_i\|}. \quad (12)$$

The Kalman gain and Joseph-form covariance update are

$$\mathbf{S}^k = \mathbf{H}^k \mathbf{P}^{k|k-1} (\mathbf{H}^k)^\top + \mathbf{R}, \quad (13)$$

$$\mathbf{K}^k = \mathbf{P}^{k|k-1} (\mathbf{H}^k)^\top (\mathbf{S}^k)^{-1}, \quad (14)$$

$$\hat{\mathbf{x}}^{k|k} = \hat{\mathbf{x}}^{k|k-1} + \mathbf{K}^k (\mathbf{r}^k - \mathbf{h}(\hat{\mathbf{p}}^{k|k-1})), \quad (15)$$

$$\mathbf{P}^{k|k} = (\mathbf{I} - \mathbf{K}^k \mathbf{H}^k) \mathbf{P}^{k|k-1} (\mathbf{I} - \mathbf{K}^k \mathbf{H}^k)^\top + \mathbf{K}^k \mathbf{R} (\mathbf{K}^k)^\top. \quad (16)$$

We use the symmetric Joseph form for numerical stability against floating-point loss of positive-semidefiniteness.

4 Experimental Setup

4.1 Sensor Geometries

Six (N , geometry) cells are evaluated, designed to span favorable and degraded configurations at each N :

Table 1: Sensor configurations evaluated. Coordinates in metres, arena $10 \text{ km} \times 10 \text{ km}$.

Cell	Description	Sensor positions
N2_wide	widely separated baseline	(0, 0), (10000, 10000)
N2_close	close pair on one edge	(0, 5000), (500, 5000)
N3_triangle	equilateral around centre	three points on a circle, $r=5000$
N3_collinear	sensors on the line $y=0$	(1000, 0), (5000, 0), (9000, 0)
N4_square	arena corners	(0, 0), (10000, 0), (10000, 10000), (0, 10000)
N4_cluster	corner cluster, 500 m	(0, 0), (500, 0), (0, 500), (500, 500)

4.2 Monte Carlo Protocol

For each cell we run 100 independent trials, each with a fresh noise realization (seed $1000 + i$ for trial i). The ground-truth trajectory and sensor positions are identical across trials; only the noise samples $\{n_i^k\}$ differ. Per-trial position errors $e^k = \|\hat{\mathbf{p}}^k - \mathbf{p}^k\|$ are recorded for both the per-step LSE estimator and the EKF. RMSE is computed in three buckets:

Overall. Steps 5 through $K - 1$ (skipping the initial filter warmup transient).

Smooth. Overall mask intersected with the complement of a 4-step turn window after each heading change.

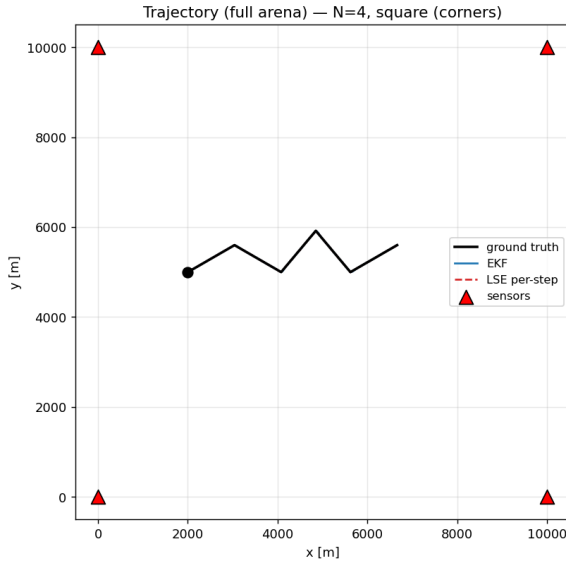
Turn. The union of the 4-step turn windows after each of the four heading changes.

This decomposition is essential: the overall RMSE blends two qualitatively different regimes and obscures the noise-floor performance.

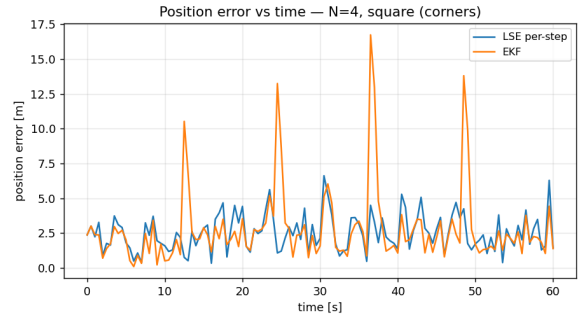
5 Results

5.1 Baseline ($N=4$ Square)

Fig. 1 shows the canonical run on the $N=4$ square cell with seed 42. The full-arena view confirms that the LSE per-step and EKF trajectories both follow the ground-truth zig-zag. The error vs. time plot clearly shows the four spikes at the heading changes and the otherwise flat baseline.



(a) Trajectory in the full 10×10 km arena.

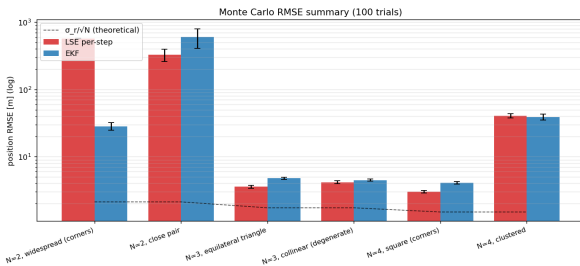


(b) Position error vs. time (single deterministic run).

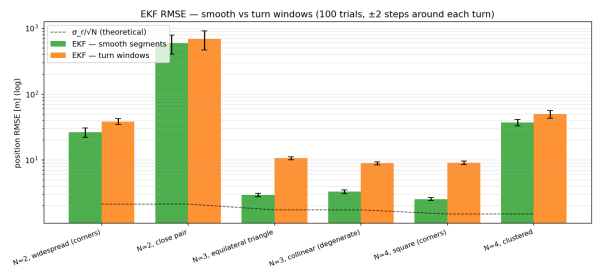
Figure 1: Baseline run on the $N=4$ square geometry, seed 42, $\sigma_a = 10 \text{ m/s}^2$.

5.2 Geometry Sweep — Monte Carlo Summary

Table 2 gives the 100-trial RMSE summary for the EKF in each of the six cells, decomposed into smooth-segment and turn-window components. Fig. 2 shows the corresponding bar chart against the $\sigma_r/\sqrt{N}$ theoretical bound.



(a) Overall RMSE; dashed line is $\sigma_r/\sqrt{N}$.



(b) EKF smooth vs. turn-window decomposition.

Figure 2: Monte Carlo RMSE summary across the six cells (100 trials each).

Table 2: EKF position RMSE, 100-trial Monte Carlo, mean $\pm$ standard deviation across trials. “Smooth” excludes the warmup and a 4-step window after each of the four heading changes; “Turn” is the union of those windows.

Cell	Overall [m]	Smooth [m]	Turn [m]
N=2, widespread (corners)	28.37 ± 3.57	26.33 ± 4.26	38.29 ± 3.91
N=2, close pair	605.00 ± 194.11	589.75 ± 190.11	691.44 ± 221.63
N=3, equilateral triangle	4.76 ± 0.18	2.92 ± 0.17	10.54 ± 0.57
N=3, collinear	4.48 ± 0.18	3.28 ± 0.22	8.84 ± 0.46
N=4, square (corners)	4.10 ± 0.17	2.52 ± 0.13	9.07 ± 0.51
N=4, clustered	38.95 ± 3.84	36.90 ± 3.92	49.55 ± 6.60

The full per-cell error-vs-time panel grid (median and inter-quartile range shading) is reproduced in Fig. 3; it shows the same story at finer time resolution.

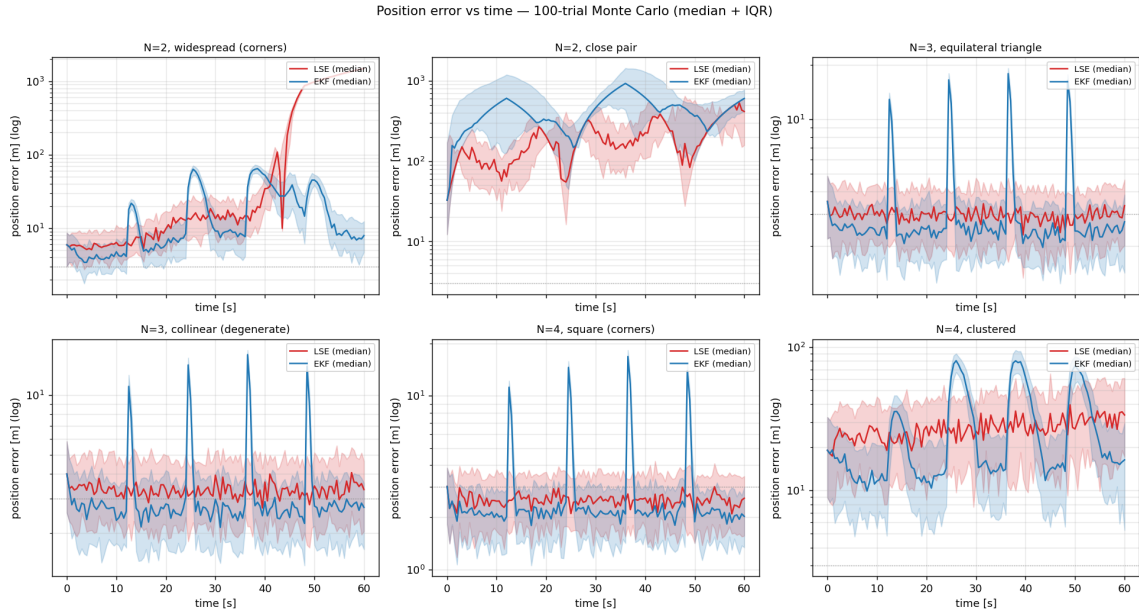


Figure 3: Position error versus time for each cell (100-trial Monte Carlo, median curves and inter-quartile range). Both estimators on log scale.

5.3 Smooth vs. Turn-Window Decomposition

The smooth-segment column of Table 2 confirms the expected $\sigma_r/\sqrt{N}$ scaling for the well-conditioned cells: the $N=4$ square cell achieves 2.52m smooth-segment RMSE against a theoretical bound of 1.5m, i.e. a factor of 1.7. The remaining gap is finite-filter overhead: the $\mathbf{P}^{k|k}$ never reaches the asymptotic floor because the four heading changes inject process noise at regular intervals.

The turn-window column tells a different story. At $N=4$ square the EKF RMSE during the post-turn transient is 9.07m, more than three times the smooth-segment number and approximately the prediction-error magnitude expected from a single-step lag: $\|\mathbf{v}\| \cdot T = 100 \cdot 0.5 = 50$ m peak, washing out over four steps to an averaged ~ 10 m. The turn-window error is therefore not random measurement noise; it is deterministic prediction lag induced by a model mismatch (the CV model does not know about the heading changes).

5.4 Collinear Stress Test

The Monte Carlo summary in Table 2 shows the $N=3$ collinear cell tracking within ~ 1 m of the $N=3$ equilateral triangle on every metric. This is unexpected: for sensors on the line $y = 0$ the range measurements are invariant under the reflection $(x, y) \mapsto (x, -y)$, so any candidate solution has a mirror-image twin. Why does the LSE/EKF pipeline not flip back and forth between them?

Because the LSE warm-start prior $\mathbf{p}_{\text{prior}} = (3000, 4000)$ m is on the same side of the sensor line as the true trajectory, and the iterative Gauss-Newton converges to whichever side it is initialised on. Once the EKF is seeded, the velocity finite-difference $\hat{\mathbf{v}}_0 = (\hat{\mathbf{p}}_1 - \hat{\mathbf{p}}_0)/T$ cements the correct branch, and there is no measurement-side information that can distinguish the two solutions.

We confirm this experimentally by running the same configuration with three priors:

Table 3: Collinear $N=3$ stress test — effect of LSE warm-start prior (single deterministic run, seed 42).

Prior position	LSE RMSE [m]	EKF RMSE [m]
(3000, 4000) above (correct side)	3.89	4.31
(3000, -4000) below (mirror image)	10 763.05	10 763.11
(5000, 0) on the sensor line	5 640.96	5 640.94

The wrong-side prior produces an RMSE of about 10.7 km, which is essentially twice the average y -coordinate of the trajectory: the EKF settles on a perfect mirror-image track and never escapes. The on-line prior produces ~ 5.6 km, half that number, because the LSE picks each side stochastically. The default configuration’s apparent robustness is therefore a feature of the prior, not of the geometry: a real deployment would need either an unambiguous prior (a fourth measurement, an asymmetric placement, or external information) or a multi-hypothesis estimator.

5.5 Process-Noise Sensitivity

Holding the geometry fixed at $N=4$ square, we sweep $\sigma_a \in \{1, 5, 10, 50, 200\}$ m/s² with 30 Monte Carlo trials per setting. Table 4 and Fig. 5 report the smooth-segment and turn-window RMSE.

Table 4: σ_a sensitivity at $N=4$ square (30-trial MC). The smooth-segment RMSE has a minimum near $\sigma_a = 10$ m/s²; the turn-window RMSE is monotone-decreasing in σ_a .

σ_a [m/s ²]	Smooth RMSE [m]	Turn RMSE [m]
1.0	22.68 $\pm$ 0.21	49.42 $\pm$ 0.42
5.0	2.72 $\pm$ 0.16	17.11 $\pm$ 0.46
10.0	2.55 $\pm$ 0.16	8.98 $\pm$ 0.46
50.0	2.93 $\pm$ 0.16	3.16 $\pm$ 0.41
200.0	3.02 $\pm$ 0.17	2.94 $\pm$ 0.38

The trade-off is monotonic and resolves the choice of σ_a as a single-axis design decision: at $\sigma_a = 1$ m/s² the filter cannot keep up with the maneuvers and accumulates > 20 m error even on smooth segments; at $\sigma_a = 200$ m/s² it essentially reduces to an instantaneous LSE plus a small Bayesian smoothing prior. No single σ_a minimizes both columns; if turn responsiveness is preferred over smooth-segment precision, $\sigma_a \in [50, 100]$ would be a better operating point than the default.

N=3 collinear sensors — EKF locks onto whichever side the prior picks
(crosses are LSE warm-start positions)

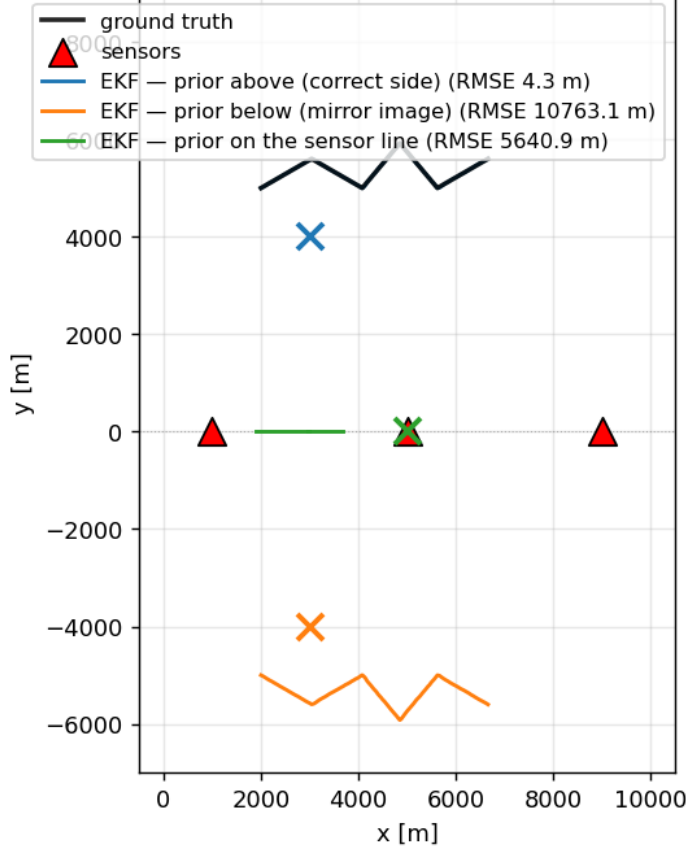


Figure 4: Collinear $N=3$: the EKF tracks the same side of the sensor line the prior was placed on; the on-line prior produces an unstable solution. Crosses mark the LSE warm-start positions.

6 Discussion

6.1 The Geometry Story

The Monte Carlo summary cleanly separates three failure modes:

1. **Pure geometric dilution** (`N4_cluster`, `N2_close`): when the unit vectors $\mathbf{u}_i$ in (6) span a small angle, the matrix $\mathbf{H}^T \mathbf{H}$ is poorly conditioned and the position covariance (8) is large. The $N=4$ clustered configuration delivers ~ 37 m smooth RMSE despite the same σ_r and the same N as the square cell — a tenfold penalty traceable entirely to GDOP.
2. **Estimator ambiguity** (`N2_close`, `N3_collinear` with wrong prior): with insufficient or symmetric geometry, the range constraints permit multiple solutions and the estimator picks one based on side information. This produces dramatic, prior-dependent failure modes that *cannot* be diagnosed from the per-cell smooth RMSE alone.
3. **Model mismatch** (every cell, during turns): the CV motion model does not know about the heading changes, so a prediction error of $\sim \|\mathbf{v}\| T = 50$ m appears at each turn and washes out over a few steps. The averaged turn-window RMSE for well-conditioned cells is ~ 9 m, four times the smooth-segment number.

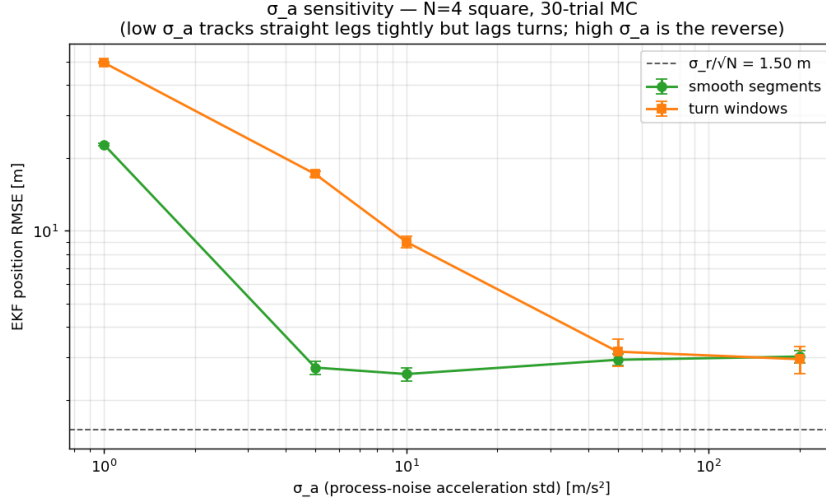


Figure 5: σ_a sensitivity sweep, $N=4$ square. Low σ_a – the filter is too rigid and accumulates lag during turns. High σ_a – the filter trusts measurements over the velocity model, so turns recover quickly but baseline noise creeps up. The default $\sigma_a = 10$ is at a reasonable knee on the smooth side.

6.2 LSE vs. EKF

A non-obvious finding is that the per-step LSE matches or slightly beats the EKF on *overall* RMSE in the well-conditioned cells. This is not an indictment of the EKF; it is a consequence of how the two estimators interact with the trajectory. The EKF’s smoothing benefit is concentrated in the smooth segments (where it gains roughly σ_r versus per-step LSE), while its lag cost is concentrated in the turn windows (where it loses ~ 5 m versus an instantaneous estimator). With four turns over a 60 s trajectory, the lag cost dominates the smooth gain in the blended RMSE. When the per-step estimator is fragile (`N2_wide`, where two-circle ambiguity produces large outliers), the EKF’s prior smooths the outliers and it wins decisively: 28 m versus 573 m.

The right metric is therefore the *decomposed* RMSE, not the blended one. On smooth segments the EKF beats per-step LSE in every well-conditioned cell; on turn windows neither estimator distinguishes itself, which is the cue for a more capable motion model.

6.3 Limitations and Future Work

Motion model. The CV model is the obvious bottleneck. An Interacting Multiple Model (IMM) filter mixing CV with a coordinated-turn (CT) model would close most of the turn-window gap. The course spec deliberately scopes this out, but the data here motivates it cleanly.

Measurement model. The TOA model assumes synchronized sensors and known emission times. TDOA relaxes the second requirement at the cost of a pairwise differencing step and a slightly higher effective noise floor; it is an alternative the spec also allows and is a natural extension.

Theoretical bound. The smooth-segment RMSE sits at $\sim 1.7\sigma_r/\sqrt{N}$ for the square cell. Computing the exact Cramér–Rao lower bound for the position estimator would quantify how far above the bound the EKF operates and is straightforward given (8).

Beyond the prior. The collinear ambiguity result suggests adding a multi-hypothesis tracker (Gaussian-Mixture or Particle Filter) for the truly ambiguous geometries. A two-hypothesis EKF that tracks both $\pm y$ branches and uses any asymmetry-breaking measurement to weight them would be a small extension.

7 Conclusion

A modest pipeline — TOA simulation, Gauss-Newton LSE initialization, and a constant-velocity EKF — recovers position to within $1.7\sigma_r/\sqrt{N}$ on smooth segments at a well-placed four-sensor geometry, and degrades predictably under three identifiable failure modes: GDOP (cluster / close-pair), estimator ambiguity (collinear / two-sensor), and motion-model mismatch (the four heading changes). The decomposed RMSE metric is essential for isolating which mode is dominant in any given cell. Of the analyses performed, the collinear stress test is the most pedagogically valuable: it shows that an apparently well-behaved geometry can hide a fundamental degeneracy that the prior is silently absorbing. The next step in the same spirit would be an IMM filter — the data already shows where it would help.

Code and Reproducibility

All experiments in this report can be reproduced from the public repository at <https://github.com/selcuko/mth407-toa-target-tracking>. Running `python main.py -no-anim` produces every figure and CSV referenced here in under a minute. Random seeds are fixed in each experiment script so results are bit-exact across runs.